

MT-331 Probability & Statistics - Final Exam Solutions

Seat No: TLE-22005

Spring Semester 2024

Question 1 (CLO-1)

(a) Correlation coefficient and regression line

Given data: $X = \{6, 2, 1, 5, 9, 12, 5\}$, $Y = \{82, 86, 43, 74, 78, 90, 78\}$.

Step 1: Compute means:

$$\bar{X} = \frac{6 + 2 + 1 + 5 + 9 + 12 + 5}{7} = 5.7143, \quad \bar{Y} = \frac{82 + 86 + 43 + 74 + 78 + 90 + 78}{7} = 75.8571$$

Step 2: Compute sums: $(X - \bar{X})$, $(Y - \bar{Y})$, $(X - \bar{X})(Y - \bar{Y})$

$$\sum (X - \bar{X})(Y - \bar{Y}) = 214.7143, \quad \sum (X - \bar{X})^2 = 87.4286, \quad \sum (Y - \bar{Y})^2 = 1432.8571$$

Step 3: Correlation coefficient:

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} = 0.607$$

Step 4: Regression line of Y on X:

$$b_1 = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2} = 2.4559, \quad b_0 = \bar{Y} - b_1 \bar{X} = 61.8235$$

$$\hat{Y} = 61.8235 + 2.4559X$$

Step 5: Predicted Y for X=5:

$$\hat{Y} = 61.8235 + 2.4559 \times 5 \approx 74.103$$

(b) First four moments

Data: $x = 0, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$, frequency $f = 1, 1, 2, 5, 10, 20, 31, 22, 11, 5, 3, 1$, total $N = 112$.

Step 1: Compute sums:

$$\sum fx = 1118, \quad \sum fx^2 = 11606, \quad \sum fx^3 = 123854, \quad \sum fx^4 = 1356278$$

Step 2: First four moments about origin:

$$\mu'_1 = 9.9821, \quad \mu'_2 = 103.625, \quad \mu'_3 = 1105.8393, \quad \mu'_4 = 12109.625$$

Question 2 (CLO-2)

(a) Coin flipping probabilities

Third head on seventh flip (Negative Binomial):

$$P = \binom{6}{2} (1/2)^3 (1/2)^4 = \frac{15}{128} \approx 0.1172$$

First head on fourth flip:

$$P = (1/2)^3 (1/2) = \frac{1}{16} = 0.0625$$

(b) Continuous random variable

Density: $f(x) = \frac{2(1+x)}{27}$, $2 \leq x \leq 5$

$$P(X < 4) = \int_2^4 \frac{2(1+x)}{27} dx = 16/27 \approx 0.5926$$

$$P(3 \leq X < 4) = \int_3^4 f(x) dx = 1/3 \approx 0.3333$$

CDF: $F(x) = \frac{x^2+2x-8}{27}$, check $P(3 \leq X < 4) = F(4) - F(3) = 1/3$

Question 3 (CLO-2)

(a) Time-to-failure

Density: $f(x) = \frac{1}{2000}e^{-x/2000}$, $x \geq 0$
CDF: $F(x) = 1 - e^{-x/2000}$

$$P(X > 1000) = e^{-1000/2000} \approx 0.6065$$

$$P(X < 2000) = 1 - e^{-1} \approx 0.6321$$

(b) Poisson accidents, $\lambda = 3$

$$P(X = 5) = \frac{3^5 e^{-3}}{5!} \approx 0.1008$$

$$P(X < 3) = P(0) + P(1) + P(2) \approx 0.4232$$

$$P(X \geq 2) = 1 - P(0) - P(1) \approx 0.8009$$

Question 4 (CLO-2)

(a) Truck blowouts, $p = 0.25, n = 15$

$$P(3 \leq X \leq 6) \approx 0.7073$$

$$P(X < 4) \approx 0.4613$$

$$P(X > 5) \approx 0.1484$$

(b) Fluorescent lights, $p = 0.9, n = 20$

$$P(X = 18) \approx 0.2852$$

$$P(X \geq 15) \approx 0.9887$$

$$P(\text{at least one fails}) \approx 0.8784$$

Question 5 (CLO-2)

(a) Soft-drink machine, $\mu = 200, \sigma = 15$

$$P(X > 224) = P(Z > 1.6) \approx 0.0548$$

$$P(191 < X < 209) = P(-0.6 < Z < 0.6) \approx 0.4515$$

Overflow in 1000 cups ($X > 230$): $P(Z > 2) = 0.0228$, expected 23 cups

(b) Poodles weights, $\mu = 8, \sigma = 0.9$

$$P(X > 9.5) = P(Z > 1.6667) \approx 0.0478$$

$$P(X \leq 8.6) = P(Z \leq 0.6667) \approx 0.7475$$

$$P(7.3 \leq X \leq 9.1) = P(-0.7778 \leq Z \leq 1.2222) \approx 0.6708$$